

GREAT FALLS, MB, Canada

WMO: 715700

Lat: 50.459N

Lon: 96.011W

Elev: 253

StdP: 98.32

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.3	(c) -29.6	(d) -36.4	(e) 0.1	(f) -32.1	(g) -33.5	(h) 0.2	(i) -29.4	(j) 7.1	(k) -11.9	(l) 6.4	(m) -12.2	(n) 1.6	(o) 170	(p) 0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	29.1	19.4	27.4	18.5	26.0	17.6	21.3	26.5	20.2	25.4	19.1	24.0	3.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.7	24.5	18.3	13.6	23.2	17.2	12.6	22.0	63.2	26.6	59.2	25.4	55.6	23.9	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.8	(b) 6.1	(c) 5.4	(e) -35.9	(f) 32.2	(g) 2.7	(h) 1.7	(i) -37.8	(j) 33.4	(k) -39.3	(l) 34.5	(m) -40.8	(n) 35.4	(o) -42.7	(p) 36.7		
(4) 6.8	(5) 6.1	(6) 5.4	(7) -36.0	(8) 23.5	(9) 2.6	(10) 1.5	(11) -37.9	(12) 24.5	(13) -39.4	(14) 25.4	(15) -40.9	(16) 26.3	(17) -42.8	(18) 27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-15.3	-14.7	-5.5	3.7	10.9	16.9	19.8	18.7	13.9	6.2	-3.1	-12.4		
	DBStd	13.90	8.38	7.42	7.68	6.05	4.95	3.44	3.15	3.12	4.38	4.82	6.56	7.39		
	HDD10.0	3448	785	692	481	199	48	1	0	0	12	141	395	694		
	HDD18.3	5618	1043	925	738	439	233	65	21	34	144	378	644	953		
	CDD10.0	1021	0	0	1	10	77	208	303	269	130	22	1	0		
	CDD18.3	150	0	0	0	0	4	22	66	45	12	1	0	0		
	CDH23.3	1131	0	0	0	1	57	179	493	323	76	2	0	0		
	CDH26.7	228	0	0	0	0	8	26	114	68	12	0	0	0		
(14) Wind	WSAvg	2.6	2.5	2.4	2.7	2.9	2.9	2.5	2.4	2.4	2.7	2.9	2.9	2.4		
(15) Precipitation	PrecAvg	469	19	13	20	27	53	85	69	65	55	34	23	19		
	PrecMax	597	69	49	43	54	162	245	156	135	110	113	86	48		
	PrecMin	286	1	0	1	0	8	8	27	10	5	1	1	3		
	PrecStd	88	16	11	13	16	35	53	34	35	30	23	20	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.8	4.3	15.4	21.3	27.6	29.4	32.0	30.9	28.3	22.5	14.1	3.3		
		MCWB	1.1	1.5	8.7	10.7	15.1	18.7	22.0	20.3	18.8	14.3	9.2	0.8		
	2%	DB	1.0	1.5	10.0	17.9	24.9	27.1	29.5	28.5	25.5	19.1	10.2	0.7		
		MCWB	-0.2	-0.1	5.5	9.2	13.7	17.5	20.3	19.0	17.5	12.5	6.3	-0.6		
	5%	DB	-0.9	-0.8	6.9	15.3	22.0	25.4	27.6	26.7	23.3	16.0	7.6	-0.5		
		MCWB	-2.0	-2.5	3.1	7.6	12.6	16.4	19.0	18.7	16.3	10.7	4.4	-1.6		
	10%	DB	-3.4	-3.8	4.3	13.0	19.3	23.8	26.1	25.0	20.9	13.3	5.1	-2.1		
		MCWB	-4.3	-5.0	1.6	6.5	11.9	16.1	18.3	17.7	15.6	9.4	3.0	-3.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	1.7	10.5	11.2	17.8	21.5	23.7	22.2	20.6	15.0	9.4	0.9		
		MCDB	2.4	3.6	12.5	18.8	21.9	26.0	30.1	27.7	25.1	21.5	13.3	2.8		
	2%	WB	-0.1	0.1	5.7	9.8	16.0	19.4	21.9	20.8	18.9	13.2	6.6	-0.3		
		MCDB	1.1	1.4	9.8	16.3	20.3	24.2	27.1	26.3	23.1	17.9	9.5	0.7		
	5%	WB	-2.1	-2.5	3.5	8.4	14.2	18.1	20.6	19.7	17.6	11.4	4.6	-1.5		
		MCDB	-1.0	-1.0	6.4	14.1	19.9	22.9	25.8	25.0	21.9	15.4	7.1	-0.6		
	10%	WB	-4.3	-5.0	1.8	7.0	12.9	16.9	19.4	18.7	16.1	9.4	3.0	-3.1		
		MCDB	-3.4	-3.7	3.9	12.7	18.1	21.7	24.5	23.4	20.0	13.0	5.0	-2.1		
(35) Mean Daily Temperature Range	MDBR		9.8	11.2	11.3	11.0	10.7	9.9	10.1	9.9	9.3	7.2	6.3	7.5		
	5% DB	MCDBR	8.9	11.2	11.1	14.1	15.0	12.7	11.9	12.4	12.1	11.1	9.0	7.1		
		MCWBR	8.0	9.3	7.4	7.5	6.7	5.7	5.4	5.7	5.7	6.2	5.9	6.1		
	5% WB	MCDBR	8.5	10.4	10.4	12.4	11.3	10.0	10.3	10.4	10.1	10.1	8.1	6.7		
		MCWBR	7.7	8.9	7.0	7.1	6.1	5.4	5.7	5.6	5.6	6.0	5.5	5.9		
(40) Clear-Sky Solar Irradiance	taub		0.257	0.280	0.291	0.340	0.354	0.377	0.385	0.370	0.341	0.301	0.274	0.248		
	taud		2.270	2.216	2.341	2.393	2.428	2.388	2.385	2.442	2.508	2.582	2.477	2.331		
	Ebn at Noon		785	855	918	904	903	880	866	863	856	841	776	748		
	Edh at Noon		67	93	100	107	109	114	113	102	86	66	56	55		
(44) All-Sky Solar Radiation	RadAvg		1.20	2.35	3.62	5.02	5.39	5.65	5.92	4.91	3.51	2.03	1.09	0.87		
	RadStd		0.12	0.18	0.26	0.37	0.43	0.46	0.31	0.37	0.32	0.26	0.14	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)		N/A	N/A	N/A	N/A	N/A	-0.68	N/A	N/A	N/A	N/A

Nomenclature: See separate page